

Weekly Problem 3

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Exercise 1

Part (a)

1		$p \Rightarrow q$	
2			$p \wedge r$
3			p $\wedge E2$
4			q $\Rightarrow E1, 3$
5		$(p \wedge r) \Rightarrow q$	$\Rightarrow I2-4$

We then see that $p \Rightarrow q \vdash (p \wedge r) \Rightarrow q$ is a valid argument in Łukasiewicz logic.

Part (b)

Let $p = 1$, and $q = i$. We see that $\neg q = i$, so $\neg q \Rightarrow q = 1$. However, $p \Rightarrow q = i$. Consequently, the argument $\neg q \Rightarrow q \vdash p \Rightarrow q$ is an invalid argument in Łukasiewicz logic.